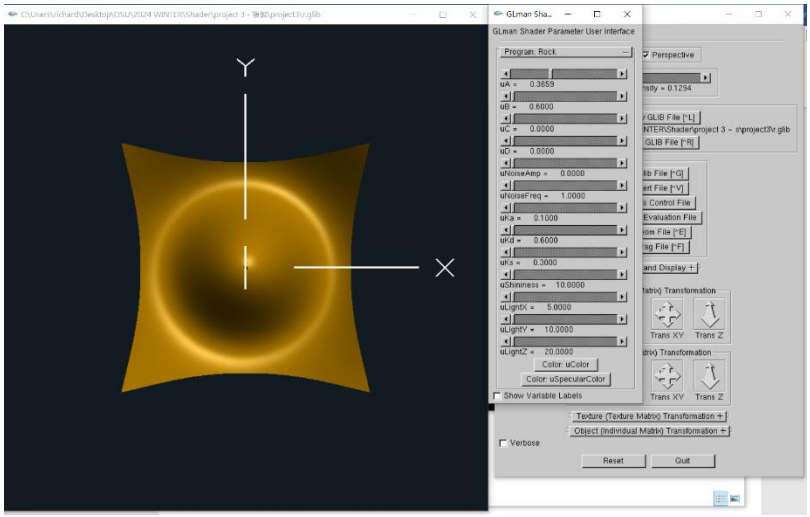
Project #3

Displacement Mapping, Bump Mapping, and Lighting

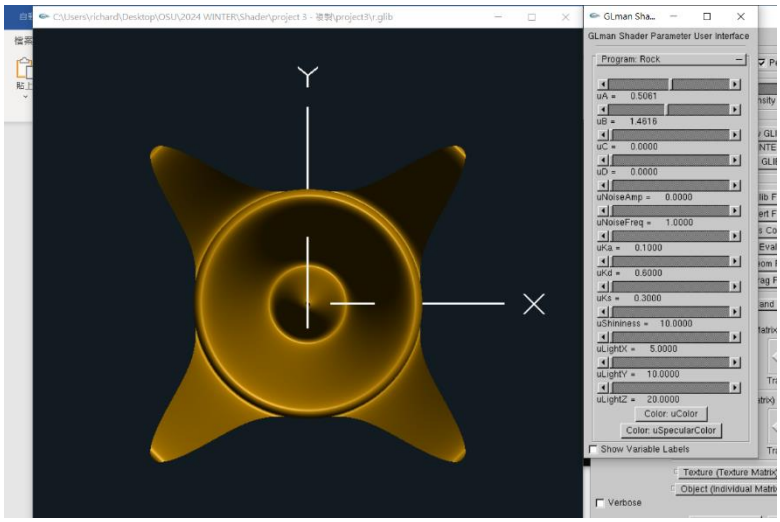
Yu-Xian Lai
laiyux@oregonstate.edu

Use the equation: Z = A * cos(2πBr+C) * e^{-Dr} for the ripple.
Set p = gl_Vertex, then the x and y for $r^2 = x^2 + y^2$ is p.x and p.y, so we can get the cross normal and pass it to the fragment shader.
Add noise and then rotate the normal, then multiply it by the gl_NormalMatrix and normalize it.

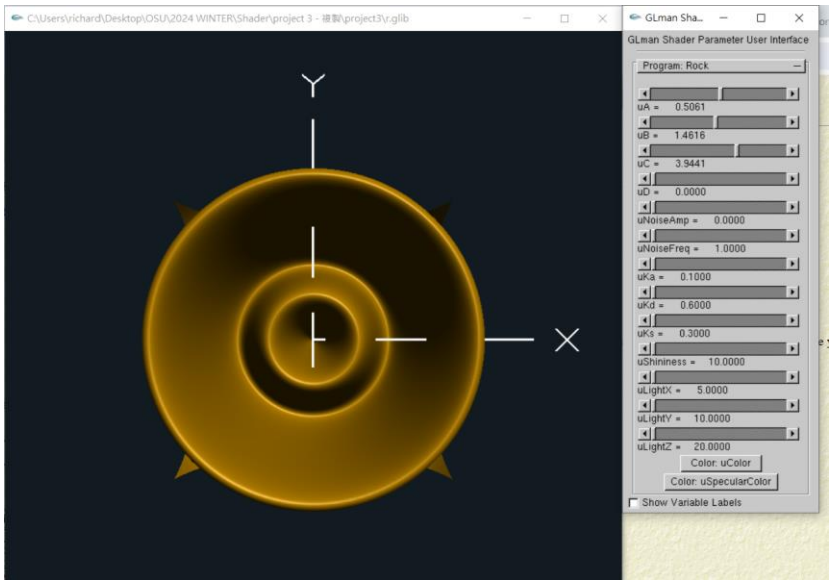
Images:
Change the value of Amplitude



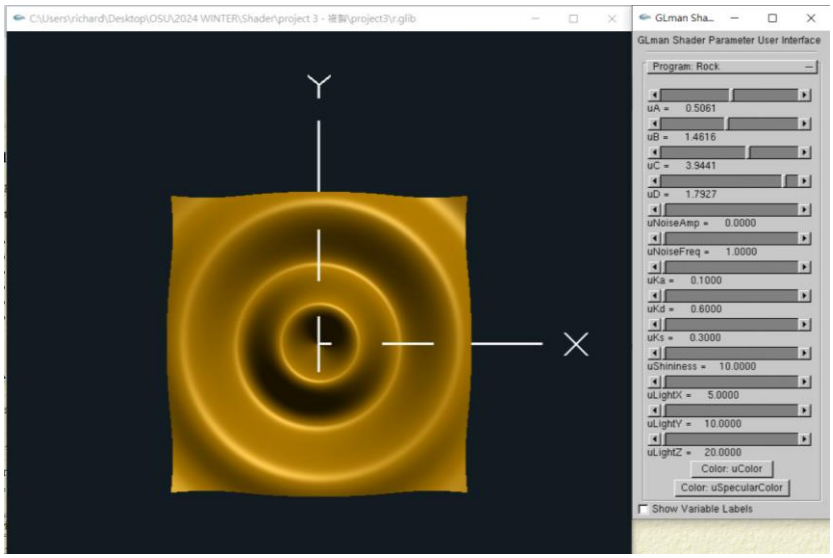
Change the value of Period



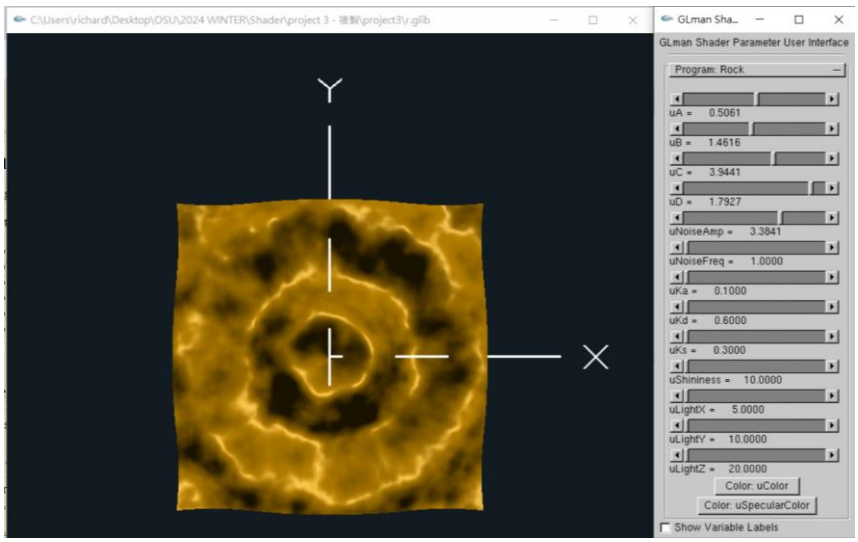
Change the value of Phase shift



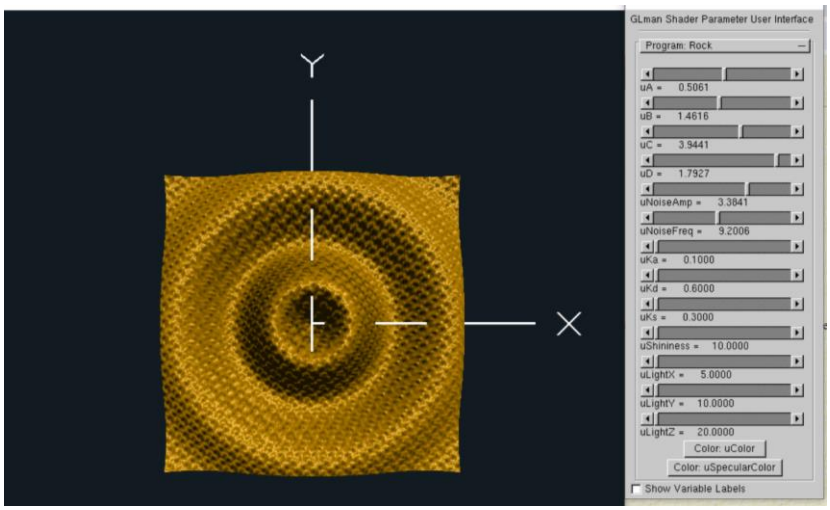
Change the value of Decay rate



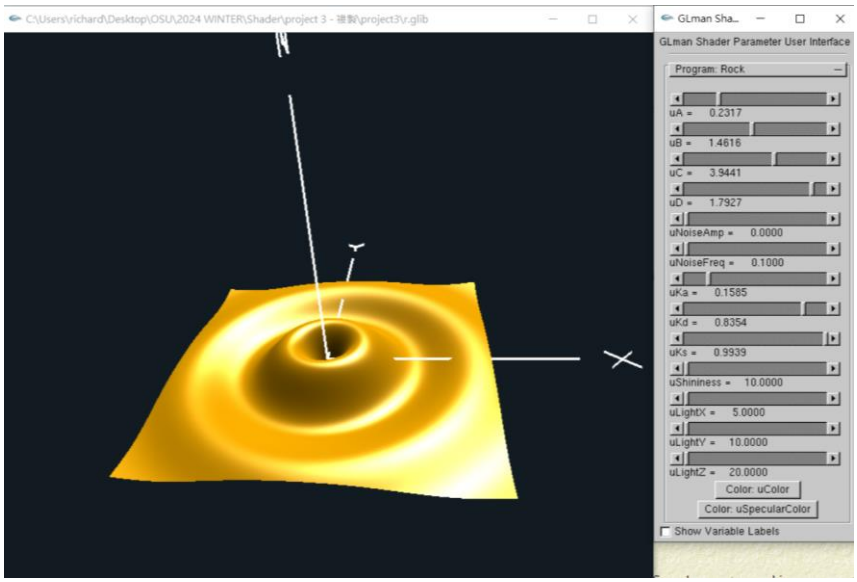
Change the value of NoiseAmp



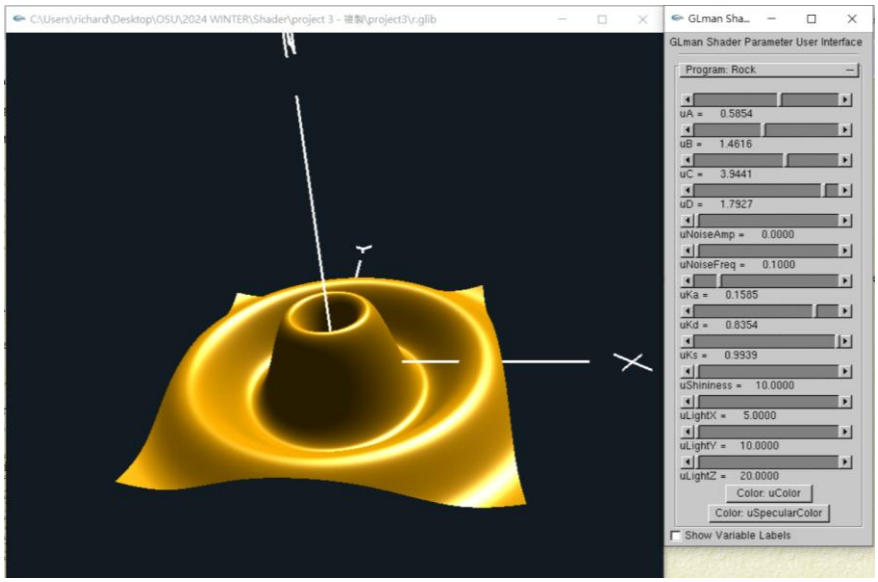
Change the value of NoiseFreq



Lighting



Bump mapping



Video link: https://media.oregonstate.edu/media/1_7v7jho1u